Laborator 6

DHCP & DNS

Scopul acestui laborator:
- Configurare DHCP (IP automat)
- Configurare DNS (nume → IP)
- Folosirea unui Web Server
- Intelegerea DHCP in alta retea (DHCP Relay)

Idee de baza:

Un PC se conecteaza automat la retea fara sa stie nimic, primeste automat configuratia si poate accesa un serviciu folosind un nume.

a) Automatizare: PC-ul Nu mai are nevoie de configurarea manuala.
b) Utilizatorul nu trebuie sa stie IP-ul, scrie numele.
c) Colaborare intre servicii: DHCP - da IP; DNS – traduce nume; router – conecteaza retelele
d) Serverele nu sunt in aceeasi retea, exista router si separare de servicii

Teorie generala:

DHCP (Dynamic Host Configuration Protocol):

- Ofera automat IP Address, Subnet Mask, Default Gateway si DNS Server
- Fara DHCP: configurare manuala pe fiecare PC, risc de erori, timp mai mare
- Functionare: PC -> „Am nevoie de IP"; DHCP Server -> „Uite IP-ul"

DHCP Realy:

- Problema: DHCP functioneaza doar in aceeasi retela
- Solutie: Routerul trimite cererea mai departe prin *ip helper-address*

DNS (Domain Name System):

- Transforma www. Abc.com -> 192.168.x.x
- Se vor retine nume, nu IP-uri

1. Topologie si schema logica:

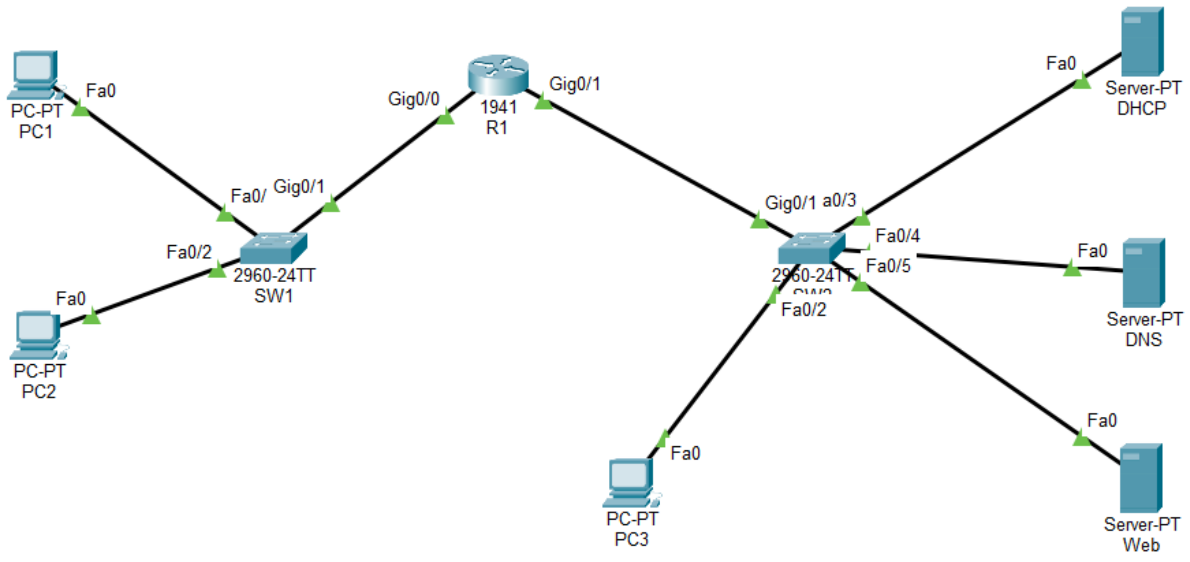
Dispozitive:

- **2 Switch-uri**
- **1 Router**
- **3 PC-uri**
- **1 DNS Server**
- **1 DHCP Server**
- **1 Web Server**

Avem 2 retele diferite plus un router intre ele.

Conexiuni:

De la	**La**	**Porturi**	**Cablu (tip)**
PC1	SW1	Fa0 -> Fa0/1	Straight
PC2	SW1	Fa0 -> Fa0/2	Straight
SW1	R1	Gi0/1 → Gi0/0	Straight
R1	SW2	Gi0/1 → Gi0/1	Straight
SW2	PC3	Fa0/2 -> Fa0	Straight
SW2	DHCP	Fa0/3 -> Fa0	Straight
SW2	DNS	Fa0/4 -> Fa0	Straight
SW2	WEB	Fa0/5 -> Fa0	Straight



2. Plan de adresare

Retea	Componente	IP
LAN 1	PC1	DHCP
	PC2	DHCP
	SW1	-
	R1	192.168.10.1
LAN 2	PC3	DHCP
	R1	192.168.20.1
	DHCP	192.168.20.100
	DNS	192.168.20.101
	WEB	192.168.20.102

3. Configurare PC-uri pe DHCP

Pe fiecare PC:

Desktop → IP Configuration -> Selectati DHCP

4. Configurarea R1

Interfata LAN 1: defineste gateway-ul pt PC-uri

```
Router(config)#interface g0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
```

Interfata LAN2: conecteaza serverele

```
Router(config-if)#interface g0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
```

DHCP Relay: trimite cererile DHCP la server

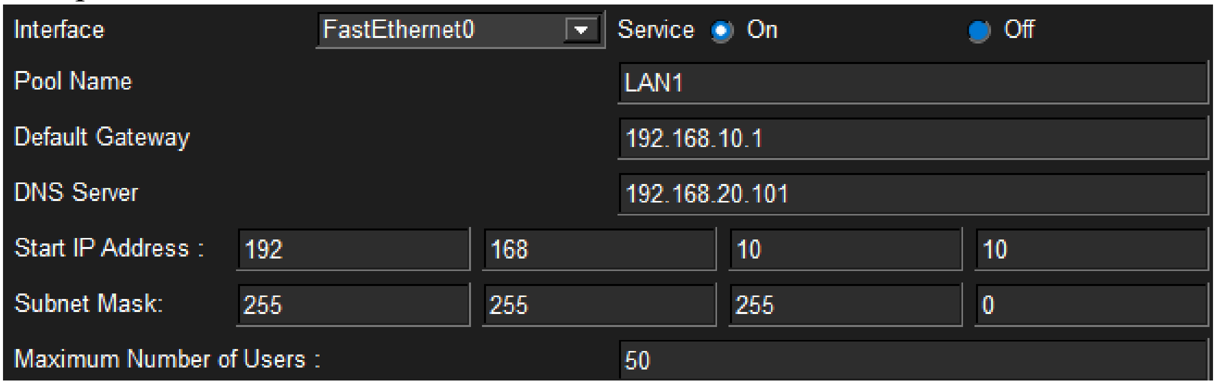
```
Router(config-if)#interface g0/0
Router(config-if)#ip helper-address 192.168.20.100
```

5. Configurarea DHCP Server

Desktop -> IP Configuration:

DHCP Service:

Services -> DHCP -> ON
Completati:



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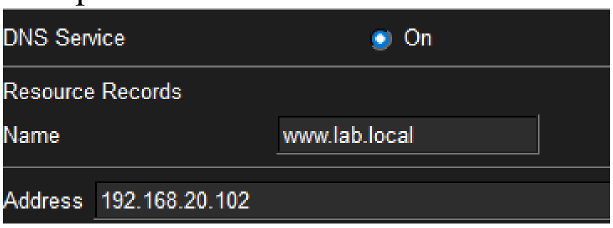
6. Configurarea DNS Server

Desktop -> IP Configuration

DNS Service:
Services -> DNS -> ON

Completati:



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7. Configurarea WEB Server

Desktop -> IP Configuration

HTTP:

Services -> HTTP -> ON

8. Verificarea configuratiei

PC1:

```
C:\>ping 192.168.10.1

Pinging 192.168.10.1 with 32 bytes of data:

Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.20.102

Pinging 192.168.20.102 with 32 bytes of data:

Reply from 192.168.20.102: bytes=32 time<1ms TTL=127
Reply from 192.168.20.102: bytes=32 time<1ms TTL=127
Reply from 192.168.20.102: bytes=32 time=5ms TTL=127
Reply from 192.168.20.102: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>ping www.lab.local

Pinging 192.168.20.102 with 32 bytes of data:

Reply from 192.168.20.102: bytes=32 time<1ms TTL=127
Reply from 192.168.20.102: bytes=32 time<1ms TTL=127
Reply from 192.168.20.102: bytes=32 time<1ms TTL=127
Reply from 192.168.20.102: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.102:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

R1:

```
Router>ping 192.168.10.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.10.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/3/8 ms

Router> ping 192.168.20.102

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.102, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms

Router>ping www.lab.local

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.20.102, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/0/1 ms
```

PC1 -> Desktop -> Web Browser